FEPrepare README file

Please cite this as Zavitsanou et al, JCIM, 2021

This folder should contain:

- Two subfolders, “complex” and “solvent”.
- Six renamed files (PDB, RTF, PRM) for the parent and the mutated ligand (reference.pdb, reference.rtf, reference.prm, mutant.pdb, mutant.rtf, mutant.prm).
- Two hybrid files for both parent and mutated ligand following the dual topology implementation of NAMD, named as “ligand.pdb” and “ligand.rtf”.
- The renamed PRM file after the creation of the two hybrid files, “updated.prm”.
- The “complex.pdb” file.
- The “fep.tcl” script that NAMD needs to run the simulation.
- A file with the OPLS-AA parameters of proteins as downloaded from http://zarbi.chem.yale.edu/oplsaam.html (par_opls_aam.inp).
- A file with the OPLS-AA topology of proteins as downloaded from http://zarbi.chem.yale.edu/oplsaam.html (top_opls_aam.inp).

The “complex” folder contains the following files:

- The PDB files for each of the protein chains and the ligand (in the case of the example “chainA.pdb” and “chainB.pdb” for the protein and “chainX.pdb” for the ligand).
- The files required to run VMD, “psfgen”, and “VMD_prepare_complex_after_gui_autopsf” or VMD_prepare_complex_after_gui_autopsf_ionized” in case the user requested 150mM NaCl salt addition.
- The files that VMD provides as output, i.e. “complex_wb.log”, “complex_wb.pdb”, “complex_wb.psf”, “psf-complex.psf”, “psf-complex.pdb”, which are required for VMD to generate the final files for the FEP calculation.
- Files named “ionized.pdb”, “ionized.fep”, “ionized.psf” as generated by VMD.
- Files named “ionized_complex.pdb” and “ionized_complex.fep”, which contain the updated ionized files, after implementing the FEPrepare NAMD/FEP Workflow (see main text for details).
- Two text files: a) “min-max_center”, which provides the cell basis vectors and the cell origin to be included in the NVT equilibration input file, and b) “vmd_log”, which describes all the steps performed by VMD.
- A file named “complex-input-files.tar.gz” which provides you with the configuration files needed to run NAMD.

The “solvent” folder contains the following files:

- The files used to run VMD, “psfgen_solv”, and “VMD_prepare_ligand_after_gui_autopsf” or

VMD_prepare_ligand_after_gui_autopsf_ionized” in case the user requested 150mM NaCl salt addition.
- The files that VMD gives as an output, “ligand_wb.log”, “ligand_wb.pdb”, “ligand_wb.psf”, “psf-solvated.psf”, “psf-solvated.pdb”, which are required for VMD to generate the final files for the FEP calculation.
- Files named “ionized.pdb”, “ionized.fep”, “ionized.psf” as generated by VMD.
- Files named “ionized_solvent.pdb” and “ionized_solvent.fep” which contain the updated ionized files, after implementing the FEPrepare
- NAMD/FEP Workflow (see main text for details).
- Two text files: a) “min-max_center”, which provides the cell basis vectors and the cell origin to be included in the NVT equilibration input file, and b) “vmd_log”, which describes all the steps performed by VMD.
- A file named “solvent-input-files.tar.gz” which provides you with the configuration files needed to run NAMD

These files are provided to the user so that they have full control of the FEPrepare workflow and VMD-generated files.

The actual files needed for the NAMD/FEP calculation are “ionized_complex.pdb”, “ionized_complex.fep”, “ionized_solvent.pdb”, “ionized_solvent.fep”, “ionized.psf”.

Your data will be deleted after two days. No other user has access to your data.

In case you do not see all the files mentioned above, there was probably an error during the calculation. Please check that you have provided the correct input or contact us at zavitsanoustamatia@gmail.com or zcournia@bioacademy.gr.